

QuestionsLab1.3.pdf



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Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
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Questions Lab1-session3

Estat	Acabat
Començat el	dimecres, 29 d'octubre 2025, 15:04
Completat el	dimecres, 29 d'octubre 2025, 15:13
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Pregunta 1

Completa

Puntuació 1,00
sobre 1,00

 Marca la
pregunta

About the meaning and calculation of the parallel fraction, which of the following statements are true regarding the analysis done in the laboratory with the five different parallel versions of 3DFFT?:

- ☐ The parallel fraction was significantly improved along each of the different versions thanks to the increment in the number of generated tasks, i.e the increment of the available parallelism.
- ☒ The parallel fraction was mainly improved thanks to the parallelization of functions not parallelized in the first versions, and in particular, when passing from version 2 to version 3.
- ☐ An approximation of the parallel fraction can be calculated using the Paraver tool, by just finding out the proportion of the duration of the parallel part in the timeline over the duration of the sequential part in the timeline in the visualized trace, when using more than one thread.
- ☐ An approximation of the parallel fraction can be calculated using the Paraver tool, by just finding out the proportion of the duration of the parallel part in the timeline over the whole duration of the program in the visualized trace, when using more than one thread.
- ☒ An approximation of the parallel fraction can be calculated using the Paraver tool, by just finding out the proportion of the duration of the parallel part in the timeline over the whole duration of the program in the visualized trace, when using only one thread.



do your thing

WUOLAH

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00 Marca la pregunta

The reasons detected about the poor scalability achieved with the first version of the OpenMP 3dfft parallel are, once we have analyzed the model factor tables and paraver views (one or more answers may be true).

- ☐ Load imbalance of the program, as not all the generated tasks have the same amount of work to do, leading to wasting of resources.
- ☐ The overhead generated when synchronizing tasks after the execution of each taskloop which is independent of the number of threads used for the execution
- ☒ The overhead generated when synchronizing tasks after the execution of each taskloop which increases with the number of threads used for the execution
- ☒ The fine grain granularity of the tasks generate a huge number of tasks which impacts directly on the task creation overhead.
- ☐ A coarser grain granularity of the tasks impacts negatively on the task synchronization overhead

Pregunta 3

Completa

Puntuació 1,00
sobre 1,00 Marca la pregunta

The final improved version of the OpenMP 3dfft parallel program was obtained after the following optimizations (one or more answers may be true):

- ☐ Incrementing the number of generated tasks which helped to add more opportunities to parallelism.
- ☒ Decrementing the total number of taskloops effectively executed in the program which lead to a lesser number of generated tasks.
- ☒ Augmenting the parallel fraction and incrementing the granularity of the tasks to reduce the synchronization and task creation overheads.
- ☐ Limiting the number of threads to execute the program.
- ☐ All the statements above are true.

[Acaba la revisión](#)**WUOLAH**